



United International University

Department of Computer Science and Engineering

CSE 3411: System Analysis and Design

Class Test - 02 Total Marks: 20 Total Time: 20 mins

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Name:	ID:
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Question 1

[10]

Suppose you want to introduce a new cloud-based attendance management system to the market. Your financial analysts estimate that the project will require an initial investment of \$12,000, followed by additional investments after certain years. After 3 years, the company must reinvest 30% of the total revenue earned up to that time. After 5 years, another investment equal to 60% of the previous reinvestment will be required. The expected revenues are \$2,500, \$3,500, \$5,000 and \$6,500 in years 2, 3, 4 and 5 respectively. Determine whether the project results in a profit or a loss using cash flow analysis.

Question 2

[10]

When is interview most effective as a requirement gathering tool? Compare with any two other requirement gathering tools.



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Question 1

[10]

A startup plans to launch an AI-based customer support application. The project requires an initial investment of \$15,000, followed by additional reinvestments. At the end of year 4, the firm must reinvest 35% of the total revenue earned by that time. After year 6, another investment equal to half of the first reinvestment will be needed. The projected revenues are \$3,000, \$4,200, \$5,800, \$6,500 and \$7,500 in years 2 through 6 respectively. Using cash flow analysis, determine the profit or loss of the project.

Question 2

[10]

When is onsite observation most effective as a requirement gathering tool? Compare with any two other requirement gathering tools.



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Question 1

[10]

Suppose a company is launching a subscription-based cybersecurity platform. The business requires an initial investment of \$8,000. After 2 years, the company must reinvest 45% of the total revenue generated up to that point. After 4 years, an additional investment equal to one-third of the first reinvestment will be required. The expected revenues are \$1,800, \$2,700, \$4,000 and \$5,500 in years 1, 2, 3 and 4 respectively. Analyze the cash flows and determine whether the project earns a profit or incurs a loss.

Question 2

[10]

When is reviewing literature and documentation most effective as a requirement gathering tool? Compare with any two other requirement gathering tools.



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Question 1

[10]

A software firm plans to launch an online data-backup service. The project begins with an initial investment of \$20,000. After 3 years, the firm must reinvest 25% of the total revenue earned by then. After 5 years, an additional investment equal to 70% of the earlier reinvestment will be required. The expected revenues are \$4,000, \$5,000, \$6,500 and \$8,000 in years 2, 3, 4 and 5 respectively. Determine the overall profit or loss using cash flow analysis.

Question 2

[10]

When is a questionnaire most effective as a requirement gathering tool? Compare with any two other requirement gathering tools.